From Trajectory-Tracking to Path-Following through Rollout

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***Abstract*—Designing a path-following controller to steer the output of a nonlinear dynamical system along a desired path at a specified speed is often significantly more challenging than designing a trajectory-tracking controller, which typically relies on a time parameterization of the path. We first consider a deterministic setting and show that, given a trajectory-tracking controller that ensures an integrable squared output error norm for any initial state and set of path parameters, it is possible to construct a path-following controller with identical guarantees. The proposed approach leverages the trajectory-tracking control policy while periodically optimizing the trajectory parameters online—specifically, the initial position and speed. Therefore, the trajectory-tracking controller with fixed parameters serves as a base policy that is refined at each time step, similar to the rollout strategy in approximate dynamic programming. The approach is then extended to a stochastic setting. We consider a disturbance model that captures both a standard Wiener process and sporadic disturbances and induces stochastic state jumps occurring at a specified rate. For the stochastic case, we show that if the system dynamics are linear and the path is generated by an exosystem, the proposed path-following strategy achieves a lower average squared output error norm compared to trajectory-tracking at constant speed. To demonstrate the generality of the method, we take trajectory-tracking policies from the literature—specifically for classical unicycle and quadcopter models—and use them to construct effective path-following controllers.**

I. INTRODUCTION

Path-following and trajectory-tracking are two related formulations of control tasks that require a system's output to be close to a desired reference. In trajectory-tracking, such a reference is simply parameterized in time, implicitly adding time constraints to the problem. In turn, path-following prioritizes the minimization of the distance between the system's output and a reference path while meeting soft time constraints, such as the desired speed. Path-following finds applications in machine tooling (e.g., CNC, laser profiling) [1]–[3], robotic manipulators [4], [5], and autonomous vehicles [6], [7]. It usually requires less demanding actuation [8] and avoids corner cutting [9].

While path-following can be beneficial in many applications, it is typically much harder to design a path-following controller than a trajectory-tracking controller. In fact, for trajectory-tracking, there are several systematic methods, based on feedback linearization [10], backstepping [11], and exploiting differential flatness [12]. In turn, path-following is often pursued in an ad hoc way: each system is treated differently, and researchers often design control laws coupled with Lyapunov functions to ensure output-reference error convergence [13]. Still, one can find general approaches to path-following. A typical approach in the context of autonomous vehicles is to design path-following controllers to handle two tasks [8]: (i) reach the desired path as a function of a so-called path variable, and (ii) satisfy an additional dynamic specification, often by controlling the path variable. A prime example of the path variable is the longitudinal position of the vehicle along the path obtained through a numerical projection [6], [7], [14]. In the context of machine tooling applications, some approaches [1]–[3] aim to reduce contouring error, which is the deviation of the output from the desired contouring path [2]. Related ideas to the present paper, but in the context of time-optimal problems, are the online trajectory time scaling [15], and the choice of trajectory speed online [16], [17]. An important aspect is that path-following control laws are often much more intricate. This can be seen by comparing some of the available trajectory-tracking policies versus path-following policies for quadcopters, [18], [19] versus [21], [22], robotic manipulators, [23], [24] versus [4], [5], mobile robots, [6], [7] versus [25], [26], and underwater vehicles, [27], [28] versus [29], [30].

The present paper tackles path-following for nonlinear systems. Considering first a deterministic setting, we show that given a trajectory-tracking control policy that results in an integrable squared output error norm, we can provide a path-following controller with the same guarantees. Such a method leverages the trajectory-tracking policy while periodically optimizing the trajectory parameters, consisting of the initial starting point and speed parameters. The trajectory-tracking policy plays then the role of a base policy that is improved at every time step, as in rollout in the context of approximate dynamic programming. The choice of parameters and the sets they can belong to are design freedoms of the proposed path-following controllers. We discuss how they can be chosen to ensure desired speed specifications through examples. We then extend the ideas to a stochastic setting. A disturbance model is considered in which disturbances induce stochastic state jumps. We argue that such a disturbance method can capture both traditional Wiener process models, when the covariance of the state jumps is scaled by the inverse of the jump occurrence rate, and sporadic noise models. For the stochastic case, we show that if the system dynamics are linear, the proposed path-following strategy guarantees a lower average squared output error norm than trajectory-tracking, provided that it is bounded for the trajectory-tracking case. We illustrate the simplicity and generality of the method by taking trajectory-tracking policies for classical unicycles and quadcopters from the literature, which ensure global and almost global stability (tracking state error convergence to zero for every initial condition and trajectory parameters). We then construct well-performing path-following policies. For the unicycle example, we also consider a previously proposed path-following policy in the literature based on

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Lyapunov theory and show that the performance of our path-following controller achieves similar results. Since obtaining a Lyapunov-based path-following controller is in general a very time-consuming task that needs to be tailored to each specific system, this illustrates that our method provides a simple, general and easy to implement alternative, streamlining the path-following design process.

The present work is a follow-up to [31], [32], which consider only the more restrictive case of linear systems. In turn, in [31], [32] the control input is also optimized, besides the trajectory parameters. This leads in general to different control input policies than those obtained by borrowing the control input policy of trajectory-tracking, as we do here for the nonlinear case. The trajectory parameters considered in [31], [32] are piecewise constants speed inputs in a horizon. In [31] finite-horizon optimal control input problems are considered and in [32] infinite-horizon problems. The linear case allows one to establish strong results such as certainty equivalence and to provide explicit path-following control policies; [32] provides a result for linear systems in the absence of noise establishing that the squared output error norm asymptotically converges to zero. One of our main results, Theorem 1 below, can be seen as a generalization of this result for nonlinear systems. Moreover, Theorem 2 below can be seen as a generalization of the result in [32] that still considers linear systems but includes stochastic disturbances.

The remainder of the paper is organized as follows. We first treat the deterministic case in Section II, as it is considerably simpler than the stochastic case; this section includes both the problem formulation and main results pertaining to this case. Section III treats the stochastic case presenting also the problem formulation and the main results. Numerical examples are discussed in Section IV. Section V presents concluding remarks. The proofs of the results are given in the appendix.

II. PATH-FOLLOWING FOR DETERMINISTIC SYSTEMS

This section starts by introducing the dynamical model and path in Section II-A. Section II-B discusses assumptions on the existence of a trajectory-tracking controller. Section II-C presents the proposed path-following policy, which has some degrees of freedom encapsulated by parameters and sets in which they can lie. The selection and assumptions on these sets are discussed in Section II-D. Section II-E provides the asymptotic error convergence result for path-following.

A. Problem formulation

Consider a continuous-time nonlinear system

$$\begin{aligned} \dot{x}(t) &= f(x(t), u(t)), \quad x(\underline{t}) = \xi, \\ z(t) &= h(x(t)), \end{aligned} \tag{1}$$

where $t \in \mathbb{R}$, $t \geq \underline{t}$, $x(t) \in \mathbb{R}^n$ is the state, $u(t) \in \mathbb{R}^m$ is the input, and $z(t) \in \mathbb{R}^p$ is the output; $\underline{t} \in \mathbb{R}$ is the initial time and $\xi \in \mathbb{R}^n$ the initial state; f is assumed to be globally Lipschitz in x and u.

Consider also a path $\gamma(s) \in \mathbb{R}^p$ parameterized by $s \in \mathbb{R}$. The general objective of a path-following controller is to let the output $z(t)$ follow the spatial path. The following error $\min_{s \in \mathcal{S}} \|z(t) - \gamma(s)\|$ should therefore be as small as possible (ideally zero). Here and throughout the paper we consider the Euclidean norm. Moreover, additional soft time constraints are typically considered, such as the desired speed.

B. Trajectory-tracking

By considering a time profile for s, the path becomes a trajectory. Here we assume that the trajectory is characterized by parameters $\theta_0 = (\sigma_0, \eta_0) \in \Theta_0 \subseteq \mathbb{R}^{n_p+1}$ where $\sigma_0 \in \mathbb{R}$ is the initial s at time $\underline{t}$, and $\eta_0 \in \mathbb{R}^{n_p}$ are the speed parameters, such that

$$\dot{s}_{\theta_0}(t) = v(t, \eta_0), \quad s_{\theta_0}(\underline{t}) = \sigma_0, \quad t \geq \underline{t}. \tag{2}$$

Several examples of parameterizations are discussed in the sequel. The reference can then be written as

$$r(t, \theta_0) = \gamma(s_{\theta_0}(t)). \tag{3}$$

In the context of, e.g., autonomous vehicles, $v(t, \eta)$ can be seen as the (trajectory) speed, while s is the arc-length.

We assume that there exists $\epsilon > 0$ such that $v(t, \eta) > \epsilon$, for every t and η. Otherwise, when the velocity is zero, the problem boils down to point stabilization, and thus it is not interesting for path-following.

The following assumption is key and states that a trajectory-tracking policy is available that leads to an integrable squared output error norm for any initial condition ξ, $\underline{t}$ and any trajectory parameters $\theta \in \Theta$, for a given set Θ such that $\Theta_0 \subseteq \Theta$.

Assumption 1. A (trajectory-tracking) controller $\mu_{\text{TT}}(x(t), t, \theta)$ is available such that, for any $\xi \in \mathbb{R}^n$, $\underline{t} \in \mathbb{R}$ and any $\theta \in \Theta$, for a given Θ such that $\Theta_0 \subseteq \Theta$,

$$\int_{\underline{t}}^{\infty} \|z(t) - r(t, \theta)\|^2 dt < \infty.$$

when $u(t) = \mu_{\text{TT}}(x(t), t, \theta)$ in (1) and $r(t, \theta)$ is given by (3). Moreover, μ_{TT} depends on θ only through the reference: if $r(t, \theta) = r(t, \theta')$ for every $t \geq t'$, then $\mu_{\text{TT}}(x, t, \theta) = \mu_{\text{TT}}(x, t, \theta')$ for every x and $t \geq t'$.

□

This assumption states that the squared output error norm is integrable for any initial condition and for a set of trajectory parameters Θ that includes those for which the trajectory corresponding to trajectory-tracking is defined. The set Θ will play an important role in the definition of the path-following policy. The rationale behind it is to ensure that the following cost is bounded when the trajectory-tracking policy is used

$$J_{\text{TT}}(\xi, \underline{t}, \theta) = \int_{\underline{t}}^{\infty} \|z(t) - r(t, \theta)\|^2 dt. \tag{4}$$

This cost function will be used as a guide for path-following to adjust online the parameters θ.

C. Proposed path-following law

Traditional works on path-following in the context of autonomous vehicles consider a numerical projection of the vehicle's position onto the path to define an error that path-following should minimize [8]. Inspired by this, we propose path-following as a policy that periodically picks online both such a numerical projection (initially σ_0) and potentially the trajectory speed parameters to minimize a cost. Such a cost function takes into account the cost (4) that would be obtained with trajectory-tracking if the trajectory corresponding to the chosen parameters would not be recomputed afterward. The control policy coincides with the trajectory-tracking policy but for the optimized trajectory parameters.

Formally, at every time step $t_k = \underline{t} + k\tau$, for a given periodic $\tau \in \mathbb{R}_{>0}$ and for $k \in \mathbb{N}_0 := \mathbb{N} \cup \{0\}$ the path-following policy computes

$$\theta_k^* \in \arg\min_{\theta \in \Theta_k} J_{\text{TT}}(x(t_k), t_k, \theta) \tag{5}$$

where $\Theta_k \subseteq \Theta$, $k \in \mathbb{N}$, are sets that can, in general, be different from Θ_0 but must also be contained in Θ. The path-following policy can then be written as

$$\mu_{\text{PF}}(x(t), t) = \mu_{\text{TT}}(x(t), t, \theta_k^*), \text{ if } t \in [t_k, t_{k+1}), \quad k \in \mathbb{N}_0. \tag{6}$$

The path-following policy (5), (6) can be interpreted as a rollout policy with base policy corresponding to trajectory-tracking [33]. In fact, at each iteration, an optimization is performed, assuming that the base policy will be used afterward. However, this optimization is repeated periodically in a receding horizon fashion.

D. Planning trajectories defined by parameters $\theta_k \in \Theta_k$

The sets Θ_k, $k \in \mathbb{N}_0$, play an important role in the proposed path-following controller. As they define the set of velocity parameters of planning trajectories at every time t_k, they can be used to ensure that path-following achieves (typically soft) velocity constraints. However, to obtain output error convergence, we need an assumption on these sets. The purpose of this section is to discuss, through examples, how one can specify these sets in a reasonable way such that meaningful desired speed profiles can be met while satisfying such an assumption.

The required assumption is stated as follows. Let

$$\mathcal{T}_k(\Theta_k) := \{r(\cdot, \theta_k)|_{[t_k, \infty)} \,|\, \theta_k \in \Theta_k\}$$

be the set of possible trajectories (functions of time, not sets of points; here $r(t, \theta_k)$ denotes the trajectory planned at t_k, i.e., $s_{\theta_k}(t_k) = \sigma_k$) for all the parameters in the set Θ_k considered in the interval $[t_k, \infty)$. Let also

$$\Pi_\ell \mathcal{T}_k(\Theta_k) := \{r(\cdot, \theta_k)|_{[t_\ell, \infty)} \,|\, \theta_k \in \Theta_k\}$$

for $\ell > k$ be defined similarly but now considering the same trajectories only in the interval $[t_\ell, \infty)$.

Assumption 2. The sets $\Theta_k \subseteq \Theta$, $k \in \mathbb{N}_0$, are such that

$$\Pi_k \mathcal{T}_{k-1}(\Theta_{k-1}) \quad \subseteq \quad \mathcal{T}_k(\Theta_k), \text{ for every } k \in \mathbb{N}. \tag{7}$$

□

This assumption states that the set of trajectories considered (and optimized) by the path-following policy at a given time t_k, parameterized by $\theta_k \in \Theta_k$, must be contained in the tail of the set of trajectories considered at time $k-1$.

The mentioned examples are discussed next:

1) Piecewise constant velocity: Inspired by [31], [32], we consider the following planning trajectories implicitly considered in the optimization (5) at time t_k

$$\mathcal{T}_k(\Theta_k) := \{r(t, \theta_k) = \gamma(s_{\theta_k}(t)) | \theta_k \in \Theta_k, t \in [t_k, \infty)\},$$

where $\dot{s}_{\theta_k}(t) = v(t, \eta_k)$, $s_{\theta_k} = \sigma_k$ and

$$v(t, \eta_k) = \begin{cases} \nu_\ell, \text{ if } t_k + \kappa\ell \le t < t_k + \kappa(\ell+1), \\ \qquad\qquad \ell \in \{0, 1, \dots, L-1\}, \\ v_f, \text{ if } t \ge L\kappa, \end{cases} \tag{8}$$

where the parameters are $\eta_k = (\nu_0, \nu_1, \dots, \nu_{L-1})$ and v_f is assumed to be given; the dependence of these parameters on k is omitted here and in the other examples below. The length κ must be such that either $\frac{\kappa}{\tau}$ is an integer or $\frac{\tau}{\kappa}$ is an integer. In the latter case the velocity is allowed to change in between optimization times t_k.

It is clear that (7) holds in either case when

$$\Theta_k = \{\theta_k = (\sigma_k, \nu_0, \dots, \nu_{L-1}) \in \mathbb{R}^{L+1} | \nu_i \ge 0, \text{ for every } i\}$$

imposes no constraints. Average speed hard or soft constraints can be imposed in many ways, e.g., by making

$$\Theta_k = \{\theta_k = (\sigma_k, \nu_0, \dots, \nu_{L-1}) \in \mathbb{R}^{L+1} | 0 \le \nu_i \le \bar{\nu}, \, \forall i \\ L\underline{v}_{\text{av}} \le \sum_{i=0}^{L-1} \nu_i \le L\bar{v}_{\text{av}}, \}$$

where $\underline{v}_{\text{av}}$ and $\bar{v}_{\text{av}}$ are the lower and upper bounds on the average speed and $\bar{\nu}$ is the upper bound on the speed. By setting $\underline{v}_{\text{av}} = \bar{v}_{\text{av}}$ we impose a hard constraint on the average speed. It is clear that (7) holds if $\bar{\nu} = \infty$ but does not necessarily hold otherwise.

Note that, contrary to the linear case considered in [31], [32], the optimization in (5) might not be easy to carry out, and one might have to resort to exhaustive search. In fact, for the linear case explicit expressions can be found for the cost (4) and thus numerically efficient methods such as the gradient method can be pursued. For the nonlinear case, we therefore only consider a very limited number of parameters L.

2) Sparse parameterizations: Motivated by the aforementioned discussion, we note that we can consider sparser parameterizations. For instance

$$v(t, \eta_k) = \begin{cases} v_i + (v_f - v_i)\dfrac{t - t_k}{\Delta}, \quad t_k \le t < t_k + \Delta, \\ v_f \quad t \ge t_k + \Delta, \end{cases}$$

where $\eta_k = (v_i, \Delta)$ parametrizes a continuous velocity profile that initially varies linearly and then remains constant; v_f is

a given constant. Average speed constraints can be added by considering the related speed profile

$$v(t,\eta_k)=\begin{cases} v_1+(v_2-v_1)\dfrac{(t-t_k)}{\Delta_1}, & t_k\le t<t_k+\Delta_1,\\ v_2+(v_f-v_2)\dfrac{t-(t_k+\Delta_1)}{\Delta_2}, & \\ \qquad t_k+\Delta_1\le t<t_k+\Delta_1+\Delta_2 & \\ v_f, \quad t\ge t_k+\Delta_1+\Delta_2, & \end{cases}$$

where now $\eta_k=(v_1,v_2,\Delta_1,\Delta_2)$. In fact to ensure hard average speed constraints it suffices that $\frac{(v_1+v_2)\Delta_1+(v_2+v_f)\Delta_2}{\Delta_1+\Delta_2}=v_f$ effectively reducing the number of parameters in η_k to three. It is clear that (7) holds if $\Delta_1>\tau$ and we impose no additional constraints on these parameters other than v_1 and v_2 being positive.

3) No velocity optimization: Suppose that $\theta_k=\sigma_k$ and $v(t,\theta_k)=v_c$ at every time t, for a given desired speed v_c both for trajectory-tracking and for path-following. With this simple parameterization, we obtain a simple method that amounts to simply optimizing the start of the trajectory. It is clear that (7) holds.

E. Path-following error convergence

Let

$$J_{\text{PF}}(\xi,\underline{t})=\int_{\underline{t}}^{\infty}\|z(t)-r(t,\bar\theta(t))\|^2dt$$

where

$$\bar\theta(t)=\theta_k^*,\ \text{if } t\in[t_k,t_{k+1}),\quad t\in\mathbb{R},\quad t\ge t_0,$$

denotes the cost of the proposed path-following policy when $u(t)=\mu_{\text{PF}}(x(t),t)$. In this section we show that $J_{\text{PF}}(\xi,\underline{t})$ is bounded provided that (4) is bounded, i.e., that Assumption 1 holds. In particular this implies that the squared norm of the tracking error converges to zero. The result is stated next.

Theorem 1. Suppose that Assumptions 1, 2 hold. Then, for any initial condition ξ, $\underline{t}$, and any $\theta_0\in\Theta_0$,

$$J_{\text{PF}}(\xi,\underline{t})\le J_{\text{TT}}(\xi,\underline{t},\theta_0). \tag{9}$$

If, in addition, $\|z(t)-r(t,\bar\theta(t))\|^2$ is Lipschitz continuous on each interval $[t_k,t_{k+1})$, $k\in\mathbb{N}_0$, with a Lipschitz constant M independent of k (e.g., if the output error and its derivative remain bounded), then

$$\lim_{t\to\infty}\|z(t)-r(t,\bar\theta(t))\|^2=0. \tag{10}$$

□

The proof can be found in the appendix.

Remark 1. (online implementation) Note that often a trajectory-tracking control law is available that achieves zero-error regulation and $J_{\text{TT}}(\xi,\underline{t},\theta)$ is bounded, but a closed-form expression for $J_{\text{TT}}(\xi,\underline{t},\theta)$ is not available. Still, $J_{\text{TT}}(\xi,\underline{t},\theta)$ can be computed online by simulating the dynamics of the system for a fixed trajectory determined by the parameters θ. In such a case, the optimization (5) needs to be performed online, similar to many applications of rollout [33].

Remark 2. Note that if we judge the performance of a strategy by the cost consisting of the integral of the squared output error norm, the path-following strategy outperforms trajectory-tracking due to (9).

III. PATH-FOLLOWING FOR STOCHASTIC SYSTEMS

In this section, we propose a path-following policy for stochastic systems. In Section III-A, the stochastic dynamical model is introduced. Section III-B describes the trajectory-tracking problem and its assumptions. The proposed path-following policy is presented in Section III-C. The special case of linear systems is discussed in Section III-D. For linear systems, boundedness of the average squared output error norm is shown in Section III-E.

A. Problem formulation

In the stochastic case, we modify (1) as follows:

$$\begin{aligned}\dot x(t)&=f(x(t),u(t)),\quad x(\underline{t})=\xi,\\ x(a_\ell)&=x(a_\ell^-)+w_\ell,\\ z(t)&=h(x(t)),\end{aligned} \tag{11}$$

where $x(a_\ell^-)=\lim_{t\to t|t<a_\ell}x(t)$, and $\{w_\ell\}_{\ell\in\mathbb{N}_0}$ is a sequence of independent and identically distributed random vectors with zero mean and finite covariance $\mathbb{E}[w_\ell w_\ell^\intercal]=W$. The time intervals $\{a_{\ell+1}-a_\ell\}_{\ell\in\mathbb{N}_0}$, with $a_0=0$, are assumed to be independent and exponentially distributed with rate λ, i.e., $\text{Prob}[a_{\ell+1}-a_\ell>b]=e^{-\lambda b}$, for every $\ell\in\mathbb{N}_0$.

This disturbance model can represent Wiener processes. In fact, as explained in [34] and [35], the stochastic differential equation $dx=cxdt+d\omega$, where ω is a Wiener process can be understood as the limit as $\lambda\to\infty$ of the jump system with continuous dynamics $\dot x=cx$, and state jumps $x(s_\ell^1)=x(s_\ell^{1-})+1/\sqrt{\lambda}$, $x(s_\ell^2)=x(s_\ell^{2-})-1/\sqrt{\lambda}$ where both $s_{\ell+1}^i-s_\ell^i$, $i\in\{1,2\}$ are exponentially distributed with rate $\frac{\lambda}{2}$; this process can be shown to be equivalent to a jump system with dynamics $\dot x=cx$ and state jumps $x(a_\ell)=x(a_\ell{}^-)+d_\ell$ with $\text{Prob}[d_\ell=1/\sqrt{\lambda}]=\text{Prob}[d_\ell=-1/\sqrt{\lambda}]=\frac12$ and with $a_{\ell+1}-a_\ell$ exponentially distributed with rate λ, which can be captured by (11). As mentioned in [34], this construction applies to more general systems. Note that the model considered here can capture other cases of interest, such as the case where λ is small and events occur sporadically.

B. Trajectory-tracking

We cannot use the previous cost (4) for trajectory-tracking because it is generally unbounded under (11). Therefore, we consider the following two costs, which are potentially bounded: (i) discounted cost

$$J_{\text{TT}}^{\alpha_C}(\xi,\underline{t},\theta_0)=\int_{\underline{t}}^{\infty}\mathbb{E}[e^{-\alpha_C(t-\underline{t})}\|z(t)-r(t,\theta_0)\|^2]dt \tag{12}$$

for some discount factor $\alpha_C>0$, and (ii) the average cost

$$J_{\text{TT,av}}=\limsup_{T\to\infty}\frac{1}{T}\int_{\underline{t}}^{\underline{t}+T}\mathbb{E}[\|z(t)-r(t,\theta_0)\|^2]dt. \tag{13}$$

The following assumption is analogous to Assumption 1.

Assumption 3. A (trajectory-tracking) control policy $\mu_{\text{TT}}(x(t), t, \theta_0)$ is available such that, for any ξ, $\underline{t}$, θ_0, and $\alpha_C > 0$,

$$\int_{\underline{t}}^{\infty} \mathbb{E}[e^{-\alpha_C(t-\underline{t})}\|z(t) - r(t, \theta_0)\|^2]dt < \infty,$$

and $J_{\text{TT,av}} < \infty$ when $u(t) = \mu_{\text{TT}}(x(t), t, \theta_0)$ in (1) and $r(t, \theta_0)$ is given by (3).

C. Proposed method

As in the method proposed for the deterministic case (5), we select at each time step t_k, the trajectory parameters as follows

$$\theta_k^* \in \arg\min_{\theta \in \Theta_k} J_{\text{TT}}^{\alpha_C}(x(t_k), t_k, \theta), \tag{14}$$

Note that, unlike the deterministic case, we now use the discounted cost (12). The path-following policy can then again be written as

$$\mu_{\text{PF}}(x(t), t) = \mu_{\text{TT}}(x(t), t, \theta_k^*), \text{ if } t \in [t_k, t_{k+1}), \quad k \in \mathbb{N}_0. \tag{15}$$

D. Linear systems

In the case of linear systems, we have

$$\begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t), \quad x(\underline{t}) = \xi, \\ x(a_\ell) &= x(a_\ell^-) + w_\ell, \\ z(t) &= C_1 x(t), \end{aligned} \tag{16}$$

In this case, building upon the internal model principle [36], we can provide an explicit trajectory-tracking controller when the path is described by an exosystem

$$\begin{aligned} \frac{d}{ds}\omega(s) &= S\omega(s), \\ \gamma(s) &= H\omega(s), \end{aligned} \tag{17}$$

$\omega(s) \in \mathbb{R}^{n_\omega}$ for every $s \in \mathbb{R}$, $s \geq \sigma$, and the corresponding trajectory is obtained by setting

$$v(t, \theta_0) = v_c, \text{ for every } t \in \mathbb{R},\ t \geq \underline{t}, \tag{18}$$

for some $v_c > 0$, with $\theta_0 = (\sigma, v_c)$. As usual, it is assumed that all the eigenvalues of S are on the imaginary axis. For this, we require the following assumptions.

Assumption 4.
1. (A, B) is stabilizable, (A, C_1) is detectable, and (S, H) is detectable.
2. $m = p$ (same number of inputs and regulated outputs)
3. $\text{rank}(P_C(s)) = n + m$ for every eigenvalue s of $v_c S$ where $P_c(s) = \begin{bmatrix} sI - A & -B \\ C_1 & 0 \end{bmatrix}$ (no transmission zeros from the input to the output coincide with the eigenvalues of $v_c S$).

Under these assumptions there exist $K \in \mathbb{R}^{m \times n}$, $\Pi \in \mathbb{R}^{n \times n_\omega}$, and $\Gamma \in \mathbb{R}^{m \times n_\omega}$ such that $(A + BK)$ is Hurwitz, $A\Pi + BK\Pi + B\Gamma = v_c \Pi S$, $C_1\Pi = H$ [36], and the control law

$$u(t) = Kx(t) + \Gamma\omega(s_\theta(t)) \tag{19}$$

satisfies Assumption 3 - this can be concluded from Proposition 1 below. For trajectories with parameters θ and possibly time-varying speed, the trajectory-tracking policy is defined as $\mu_{\text{TT}}(x(t), t, \theta) = Kx(t) + \Gamma\omega(s_\theta(t))$, with the same K, Π, Γ. We need the following additional assumption on the sets Θ_k, stating that the planned trajectories return to the nominal speed v_c after one sampling period, and that constant-speed trajectories are always among the options.

Assumption 5. For every $k \in \mathbb{N}_0$ and every $\theta_k = (\sigma_k, \eta_k) \in \Theta_k$, $v(t, \eta_k) = v_c$ for every $t \geq t_k + \tau$. Moreover, for every $k \in \mathbb{N}_0$ and $\sigma \in \mathbb{R}$, Θ_k contains the parameters (σ, η_c) of the constant-speed trajectory ($v(t, \eta_c) = v_c$ for every $t \geq t_k$), and the trajectory-tracking parameters θ_0 belong to Θ_0.

This assumption holds, for instance, for the piecewise constant velocity parameterization of Section II-D with $v_f = v_c$, $L\kappa \leq \tau$, and $\nu_i = v_c$ allowed. Note that it implies Assumption 2. It is used in the following proposition, which in turn is key to define a policy such as (14) but for $\alpha_C = 0$. This policy is of interest because, as we shall establish, it allows us to reduce the average cost (13) associated with trajectory-tracking.

Proposition 1. Suppose that the dynamical system and reference are described by (16) and (17)-(18), respectively, and that Assumptions 4, 5 hold, and that the trajectory-tracking control policy is given by (19). Then, for every ξ, $k \in \mathbb{N}_0$, $\theta_k \in \Theta_k$ and $\alpha_C > 0$,

$$J_{\text{TT}}^{\alpha_C}(\xi, t_k, \theta_k) = J_{\text{TT}}^{\alpha_C, \tau}(\xi, t_k, \theta_k) + e^{-\alpha_C \tau}\frac{\lambda}{\alpha_C}\text{tr}(P_{\alpha_C} W),$$

where, for $\beta(t) = x(t) - \Pi\omega(s_{\theta_k}(t))$,

$$\begin{aligned} J_{\text{TT}}^{\alpha_C, \tau}(\xi, t_k, \theta_k) = \mathbb{E}\Big[\int_{t_k}^{t_k + \tau} e^{-\alpha_C(t - t_k)}\|z(t) - r(t, \theta_k)\|^2 dt \\ + e^{-\alpha_C \tau}\beta(t_k + \tau)^\intercal P_{\alpha_C}\beta(t_k + \tau)\Big] \end{aligned}$$

and P_{α_C} is the unique solution to

$$(A + BK - \tfrac{\alpha_C}{2}I)^\intercal P_{\alpha_C} + P_{\alpha_C}(A + BK - \tfrac{\alpha_C}{2}I) + C_1^\intercal C_1 = 0.$$

Moreover, $J_{\text{TT,av}} = \lambda\,\text{tr}(PW)$, where $P := P_0$.

□

The proof can be found in the appendix.

Note that the last term in the expression for $J_{\text{TT}}^{\alpha_C}(\xi, t_k, \theta_k)$ depends neither on ξ nor on θ_k. Thus, for every $\alpha_C > 0$, the minimizers in (14) coincide with those of $J_{\text{TT}}^{\alpha_C, \tau}$. As $\alpha_C \to 0$, this constant term converges to infinity, whereas $P_{\alpha_C} \to P$ and $J_{\text{TT}}^{\alpha_C, \tau}$ converges to the finite cost $J_{\text{TT}}^{0, \tau}$ obtained by setting $\alpha_C = 0$ in its definition. Therefore we can define the following policy, which is well posed.

$$\theta_k^* \in \arg\min_{\theta \in \Theta_k} J_{\text{TT}}^{0, \tau}(x(t_k), t_k, \theta), \tag{20}$$

$$\mu_{\text{PF}}(x(t), t) = \mu_{\text{TT}}(x(t), t, \theta_k^*), \text{ if } t \in [t_k, t_{k+1}), k \in \mathbb{N}_0 \tag{21}$$

E. Average error norm boundedness of path-following

The next result pertains to the average cost achieved by the path-following control policy (20), (21). Let

$$J_{\text{PF,av}} = \lim \sup_{T\to\infty} \frac{1}{T} \int_{\underline{t}}^{\underline{t}+T} \mathbb{E}[\|z(t) - r(t, \bar{\theta}(t))\|^2]dt$$

denote the average cost obtained with the path-following control policy (20), (21).

Theorem 2. Suppose that the dynamical system and the reference are described by (16) and (17)-(18), respectively, and that Assumptions 4, 5 hold, and that the trajectory-tracking control policy is given by (19). Then the average cost obtained by policy (20), (21) is not greater than that of trajectory-tracking, i.e.,

$$J_{\text{PF,av}} \leq J_{\text{TT,av}}.$$

□

The proof can be found in the appendix.

IV. NUMERICAL EXAMPLES

In this section, the performance of the proposed path-following controller is compared with that of a trajectory-tracking controller and an ad hoc path-following controller from the literature. Two systems are considered. The first system analyzed is a unicycle robot that moves in a plane, while the second system is a quadcopter that flies in a three-dimensional environment. Both systems are tracking a periodic path, which is assumed to be described by the Fourier series

$$\gamma(p) = a + \sum_{i=1}^{N} b_i \cos\left(\frac{2\pi}{T} ip\right) + c_i \sin\left(\frac{2\pi}{T} ip\right) \qquad (22)$$

with $a \in \mathbb{R}^d$, $b_i \in \mathbb{R}^d$, $c_i \in \mathbb{R}^d$, $i \in \{1, \ldots, N\}$, for $p \geq 0$. To consider the path over one period we can let $p \in [0, T)$. This Fourier description can be used to characterize any periodic path when $N = \infty$, but, in practice, a finite N, can be used to describe paths of interest. It is assumed that the paths are followed with constant speed $\bar{V}$, and that the start of the trajectory σ is the only parameter that is optimized (method 3 in Section II-D). Since the arclength of the path by $s(t)$ is given by $s(t) = \delta(p) = \int_0^p \|\gamma'(v)\|dv$, the path is given as $\gamma(\delta^{-1}(\sigma + \bar{V}t))$ for $t \geq 0$, for some $\sigma = \delta(\rho_0)$, $\rho_0 \in [0, T)$ defining along the path. Based on the path description in (22), appropriate references $r(t, \theta)$ are generated for both systems.

A. Unicycle robot

The dynamics of the unicycle robot are given by

$$\begin{aligned} \dot{x}(t) &= V(t)\cos(\varphi(t)) \\ \dot{y}(t) &= V(t)\sin(\varphi(t)) \\ \dot{\varphi}(t) &= \omega(t) \end{aligned} \qquad (23)$$

where $x(t), y(t) \in \mathbb{R}$ are the planar coordinates, and $\varphi(t) \in \mathbb{R}$ is the yaw angle of the robot at time $t \in [0, \infty)$. $V(t)$ and $\omega(t)$ are the control inputs.

Based on the path description in (22), a reference $r(t, \theta)$ is determined. Ideally, the reference would be defined as $r(t, \theta) = [\bar{x}(t), \bar{y}(t), \bar{\varphi}(t)]^\top$. Since the angle error $\varphi - \bar{\varphi}$ is ill-defined because angles are defined modulo 2π, we use a modified cost function. Let $\rho(t) = [\cos(\varphi(t)), \sin(\varphi(t))]^\top$ and $\bar{\rho}(t) = [\cos(\bar{\varphi}(t)), \sin(\bar{\varphi}(t))]^\top$, then $e_\rho(t) = 1 - \rho(t)^\top \bar{\rho}(t)$ can be seen as an error variable. The cost function being optimized is therefore $J(\xi, \underline{t}, \theta) = \int_{\underline{t}}^{\infty} e_x^2(t) + e_y^2(t) + e_\rho(t)dt$ where $e_x(t) = x(t) - \bar{x}(t)$ and $e_y(t) = y(t) - \bar{y}(t)$. The trajectory-tracking control law ensures that

$$\lim_{t\to\infty} e_x(t) = 0, \lim_{t\to\infty} e_y(t) = 0, \lim_{t\to\infty} e_\rho(t) = 0.$$

Details of the trajectory-tracking control law can be found in the appendix.

B. Quadcopter

The quadcopter model is taken from [18], and its dynamics are given by

$$\begin{aligned} \dot{\rho}(t) &= R(t)\nu(t) \\ \dot{\nu}(t) &= -S(\omega(t))\nu(t) + gR^\top(t)e_3 - \frac{f(t)}{m}e_3 \\ \dot{R}(t) &= R(t)S(\omega(t)) \\ J\dot{\omega}(t) &= S(J\omega(t))\omega(t) + \tau(t). \end{aligned}$$

$\rho(t) \in \mathbb{R}^3$ denotes the position of the center mass, $\nu(t) \in \mathbb{R}^3$ denotes the body-frame linear velocity, and $\omega(t) \in \mathbb{R}^3$ denotes the body-frame angular velocity. R is the rotation matrix from the fixed-body frame to the inertial frame. The control inputs are $f(t) \in \mathbb{R}$ and $\tau(t) \in \mathbb{R}^3$, and denote the total thrust and total moment vector in the body-fixed frame, respectively. Furthermore, m is the total mass, $e_3 = [0, 0, 1]^\top$, $J \in \mathbb{R}^{3\times3}$ is the inertia matrix, and S a skew-symmetric matrix, implicitly defined by $S(a)v = a \times v$, where $\times$ denotes the cross product. As the trajectory-tracking controller, we use the controller proposed in [18]. The reference path for the center of mass position $\rho(t)$ is described as in (22), and converted into the appropriate references according to the procedure described in [18].

C. Deterministic nonlinear path-following

In Figure 1a, the performance of the proposed path-following controller is compared with that of the trajectory-tracking controller. Additionally, we compare it with a path-following controller taken from the literature designated for unicycle robots and described in [13]. We initialize the robot at $x(0) = 1.5$, $y(0) = 0$, $\varphi(0) = \frac{1}{2}\pi$, and the desired path to follow is a circle. The trajectory-tracking controller chooses an entirely different route to the circle than the path-following controllers. The proposed path-following controller performs similarly to the path-following controller from the literature. This can also be seen in Figure 1c, where a measure of the tracking error of the different control strategies is visualized. The path-following controllers reduce the error to nearly zero within 3 seconds, whereas the trajectory-tracking controller takes 7 seconds to achieve a similar reduction in the path-following error.

The quadcopter paths under different control strategies are illustrated in Figure 2a. The quadcopter is initialized at rest

TABLE I: Average tracking error with stochastic disturbances, averaged over 50 Monte Carlo simulations.

	Trajectory-tracking	Path-following
Off-path initialization	17.246	1.446
On-path initialization	0.357	0.337

from $\rho(0) = [0, -1, 0]^\top$, and the desired path to track is a tilted circle. The position errors of the center of mass relative to the path are shown in Figure 2b. Although the initial state is close to the path, the trajectory-tracking controller has large initial errors and takes relatively long to settle onto the path. The path-following controller, on the other hand, reduces the tracking error significantly faster.

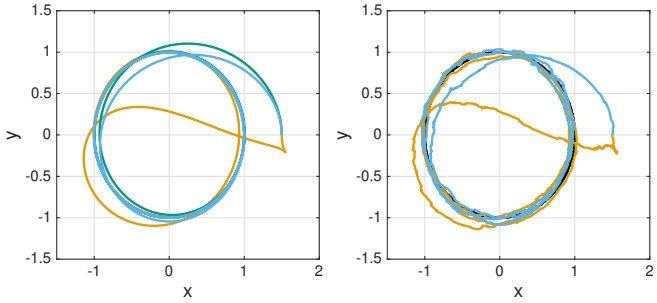
D. Stochastic nonlinear path-following

For the unicycle robot, the behavior in the presence of disturbances is shown in Figure 1b. The disturbances are characterized by $\lambda = 20$ and $W = \text{diag}([0.01, 0.01, 0.01])$. These settings imply that the average time between stochastic jumps is 50 milliseconds. The paths taken by both controllers are comparable to those in the deterministic setting shown in Figure 1a.

To assess the tracking error, we compute the average error of the first 20 seconds and then average it over 50 Monte Carlo simulations:

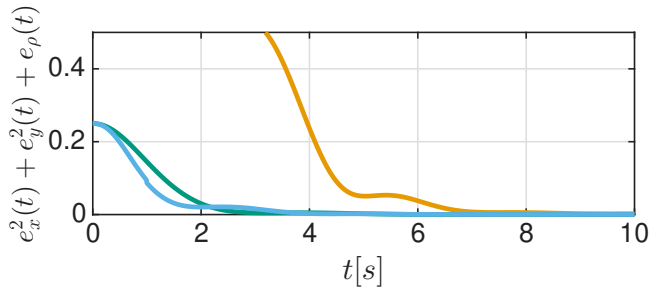
$$\bar{e} = \frac{1}{n_{\text{MC}}} \sum_{i=1}^{n_{\text{MC}}} \left[\frac{1}{T} \int_0^T e_{i,x}^2(t) + e_{i,y}^2(t) + e_{i,\rho}(t) dt \right].$$

The results are given in Table I. The proposed path-following strategy achieves an average error reduction of more than 90% compared with trajectory-tracking, given this initial (off-path) position. Furthermore, the proposed path-following controller showed improved tracking performance when the robot is initialized on the path, so the initial errors are zero. By recomputing the reference every τ seconds, the controller minimizes the path-following error based on the current state in the presence of disturbances. These results are also presented in Table I.



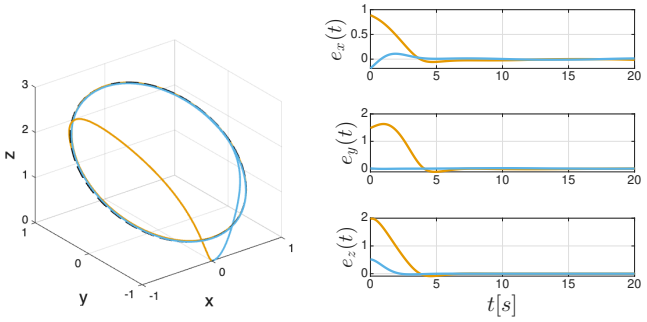
(a) Without disturbances.

(b) With disturbances.



(c) Tracking error in the deterministic setting.

Fig. 1: Comparison of trajectory-tracking (—) with the method in [13] (—), and the proposed path-following controller (—) for the unicycle robot.



(a) Path visualization.

(b) Position tracking errors.

Fig. 2: Comparison of trajectory-tracking (—) with path-following (—) for the quadcopter without disturbances.

V. CONCLUSIONS AND FUTURE WORK

In this paper, a framework is proposed to convert a trajectory-tracking controller into a path-following controller through rollout. By optimizing the path parameters, a reduced tracking error compared to ordinary trajectory-tracking is achieved. In the deterministic setting, we showed that the squared norm of the output tracking error converges to zero asymptotically. Moreover, the cost of the proposed path-following controller is upper bounded by the cost of the trajectory-tracking controller. For the stochastic setting, we showed that if the system dynamics are linear, the proposed path-following strategy guarantees a better average squared norm of the tracking error than that of trajectory-tracking. The efficacy of the proposed controller is shown in the control of a unicycle robot and of a quadcopter. In future work, we plan to tackle the stochastic setting for nonlinear systems.

APPENDIX

Proof of Theorem 1

Throughout, for $\theta \in \Theta_k$, $r(t, \theta)$ denotes the trajectory planned at time t_k, i.e., with $s_\theta(t_k) = \sigma$. Fix ξ, $\underline{t} = t_0$ and $\theta_0 \in \Theta_0$, and consider the family of control policies μ^ι, $\iota \in \mathbb{N}_0$, defined as follows. For $\iota = 0$, $\mu^0(x, t) = \mu_{\text{TT}}(x, t, \theta_0)$ for every $t \geq t_0$. For $\iota \in \mathbb{N}$, μ^ι optimizes the trajectory parameters as in (5) only at the times $t_0, \ldots, t_{\iota-1}$ and keeps the last parameters afterwards, i.e., $\mu^\iota(x, t) = \mu_{\text{TT}}(x, t, \theta_k^*)$ if $t \in [t_k, t_{k+1})$, $k \in \{0, \ldots, \iota-2\}$, and $\mu^\iota(x, t) = \mu_{\text{TT}}(x, t, \theta_{\iota-1}^*)$ if $t \geq t_{\iota-1}$. Let $\tilde{J}^\iota(\xi, \underline{t})$ denote the integral of the squared output error norm obtained with μ^ι, where the reference is the one used by μ^ι at each time. Then $\tilde{J}^0(\xi, \underline{t}) = J_{\text{TT}}(\xi, \underline{t}, \theta_0) < \infty$ by Assumption 1. Note that μ^ι coincides with μ_{PF} on $[t_0, t_\iota)$; hence the state and the error obtained with μ^ι coincide with

those of the path-following policy on $[t_0, t_\iota]$, and so do the parameters $\theta_0^*, \dots, \theta_\iota^*$.

We show that $\tilde{J}^{\iota+1}(\xi, \underline{t}) \le \tilde{J}^{\iota}(\xi, \underline{t})$ for every $\iota \in \mathbb{N}_0$. Since μ^ι and $\mu^{\iota+1}$ coincide on $[t_0, t_\iota)$, both costs share the same contribution on $[t_0, t_\iota)$ and the same state $x(t_\iota)$, and they differ only in the tail on $[t_\iota, \infty)$. For $\mu^{\iota+1}$ this tail equals $J_{\mathrm{TT}}(x(t_\iota), t_\iota, \theta_\iota^*)$. For μ^ι, $\iota \ge 1$, the tail is the cost of tracking $r(t, \theta_{\iota-1}^*)$, $t \ge t_\iota$, with $\mu_{\mathrm{TT}}(\cdot, \cdot, \theta_{\iota-1}^*)$. By Assumption 2, there exists $\theta' \in \Theta_\iota$ such that $r(t, \theta') = r(t, \theta_{\iota-1}^*)$ for every $t \ge t_\iota$ and, by the last statement in Assumption 1, the two corresponding control policies coincide for $t \ge t_\iota$. Therefore the tail of μ^ι equals $J_{\mathrm{TT}}(x(t_\iota), t_\iota, \theta')$ (for $\iota = 0$ simply take $\theta' = \theta_0 \in \Theta_0$). By (5), $J_{\mathrm{TT}}(x(t_\iota), t_\iota, \theta_\iota^*) \le J_{\mathrm{TT}}(x(t_\iota), t_\iota, \theta')$, which proves the claim. Hence

$$\int_{t_0}^{t_\iota} \|z(t) - r(t, \bar{\theta}(t))\|^2 dt \le \tilde{J}^{\iota}(\xi, \underline{t}) \le \cdots \le \tilde{J}^{0}(\xi, \underline{t}),$$

where the first inequality holds because the integrand is nonnegative and μ^ι coincides with μ_{PF} on $[t_0, t_\iota)$. Letting $\iota \to \infty$ and using the monotone convergence theorem yields (9).

To prove (10), let $g(t) := \|z(t) - r(t, \bar{\theta}(t))\|^2$, which is nonnegative and integrable by (9) and Lipschitz continuous with constant M on each interval $[t_k, t_{k+1})$. Suppose that $g(t)$ does not converge to zero. Then there exist $\epsilon > 0$ and a sequence $\{s_j\}_{j \in \mathbb{N}}$ with $s_{j+1} \ge s_j + \tau$ and $g(s_j) \ge \epsilon$. Let $\delta := \min(\frac{\epsilon}{2M}, \frac{\tau}{2})$. If $s_j \in [t_k, t_{k+1})$, then either $[s_j, s_j + \delta]$ or $[s_j - \delta, s_j]$ is contained in $[t_k, t_{k+1})$ (the former if $s_j < t_k + \frac{\tau}{2}$, the latter otherwise); call it I_j. Then $g(t) \ge \epsilon - M\delta \ge \frac{\epsilon}{2}$ for $t \in I_j$ and, since the intervals I_j do not overlap, $\int_{t_0}^{\infty} g(t)dt \ge \sum_j \frac{\epsilon\delta}{2} = \infty$, a contradiction. □

Proof of Proposition 1

Let $A_K := A + BK$ and fix k, ξ, $\theta_k \in \Theta_k$ and $\alpha_C > 0$; let $\hat{t} := t_k + \tau$ and $\beta(t) = x(t) - \Pi\omega(s_{\theta_k}(t))$. By Assumption 5, $\dot{s}_{\theta_k}(t) = v_c$ for $t \ge \hat{t}$, so that $\frac{d}{dt}\omega(s_{\theta_k}(t)) = v_c S\omega(s_{\theta_k}(t))$. Then, between jumps and for $t \ge \hat{t}$, using (19) and $A\Pi + BK\Pi + B\Gamma = v_c \Pi S$,

$$\begin{aligned}\dot{\beta}(t) &= A_K x(t) + B\Gamma\omega - v_c \Pi S\omega \\ &= A_K x(t) - A_K \Pi\omega = A_K \beta(t),\end{aligned}$$

where we omitted the argument $s_{\theta_k}(t)$ of ω. Since $\omega(s_{\theta_k}(t))$ is continuous, $\beta(a_\ell) = \beta(a_\ell^-) + w_\ell$. Moreover, $z(t) - r(t, \theta_k) = C_1 x(t) - H\omega = C_1\beta(t)$ since $C_1\Pi = H$. Thus, for $t \ge \hat{t}$,

$$\beta(t) = e^{A_K(t-\hat{t})}\beta(\hat{t}) + \sum_{\ell:\, \hat{t} < a_\ell \le t} e^{A_K(t - a_\ell)} w_\ell.$$

Due to the memoryless property of the exponential distribution, the jump times in $(\hat{t}, \infty)$ form a Poisson process with rate λ, independent of the history up to $\hat{t}$, which we denote by $\mathcal{F}_{\hat{t}}$. Since the w_ℓ are independent with zero mean and covariance W, the cross terms vanish and, by Campbell's theorem,

$$\begin{aligned}\mathbb{E}[\beta(t)\beta(t)^\intercal | \mathcal{F}_{\hat{t}}] &= e^{A_K(t-\hat{t})}\beta(\hat{t})\beta(\hat{t})^\intercal e^{A_K^\intercal(t-\hat{t})} \\ &\quad + \lambda \int_0^{t-\hat{t}} e^{A_K r} W e^{A_K^\intercal r} dr. \end{aligned} \tag{24}$$

Let $N(r) := e^{A_K^\intercal r} C_1^\intercal C_1 e^{A_K r}$. Since $A_K - \frac{\alpha_C}{2} I$ is Hurwitz, $P_{\alpha_C} = \int_0^\infty e^{-\alpha_C r} N(r) dr$ is the unique solution to the Lyapunov equation in the statement. Using (24) and interchanging the order of integration,

$$\begin{aligned}&\mathbb{E}\Big[\int_{\hat{t}}^{\infty} e^{-\alpha_C(t - t_k)} \|C_1\beta(t)\|^2 dt \,\Big|\, \mathcal{F}_{\hat{t}} \Big] \\ &= e^{-\alpha_C \tau} \Big(\beta(\hat{t})^\intercal P_{\alpha_C} \beta(\hat{t}) \\ &\quad + \lambda \int_0^\infty e^{-\alpha_C u} \int_0^u \mathrm{tr}(N(r)W) dr\, du \Big) \\ &= e^{-\alpha_C \tau} \Big(\beta(\hat{t})^\intercal P_{\alpha_C} \beta(\hat{t}) + \frac{\lambda}{\alpha_C} \mathrm{tr}(P_{\alpha_C} W) \Big),\end{aligned}$$

where we used $\int_0^\infty e^{-\alpha_C u} \int_0^u f(r) dr\, du = \frac{1}{\alpha_C} \int_0^\infty e^{-\alpha_C r} f(r) dr$. Splitting the integral in (12) at $\hat{t}$, taking expectations and using the tower property yields the expression for $J_{\mathrm{TT}}^{\alpha_C}(\xi, t_k, \theta_k)$.

For the average cost, consider trajectory-tracking with constant speed v_c from $\underline{t}$. The same arguments with $\hat{t}$ replaced by $\underline{t}$ (note that $\beta(t)$ then obeys $\dot{\beta} = A_K\beta$ between jumps for every $t \ge \underline{t}$) lead to

$$\begin{aligned}\mathbb{E}[\|z(t) - r(t, \theta_0)\|^2] &= \beta(\underline{t})^\intercal N(t - \underline{t}) \beta(\underline{t}) \\ &\quad + \lambda \int_0^{t-\underline{t}} \mathrm{tr}(N(r)W) dr.\end{aligned}$$

The first term converges to zero exponentially fast since A_K is Hurwitz, and the second converges to $\lambda\,\mathrm{tr}(PW)$ as $t \to \infty$, where $P = \int_0^\infty N(r) dr$. Since the time average of a function converging to a limit converges to the same limit, $J_{\mathrm{TT,av}} = \lambda\,\mathrm{tr}(PW)$. □

Proof of Theorem 2

Let $A_K := A + BK$, $\underline{t} = t_0$, and let $\mathcal{F}_k$ denote the history of the closed loop up to time t_k. Define $\beta_0 := \xi - \Pi\omega(\sigma_0)$, where $\theta_0 = (\sigma_0, v_c)$ are the trajectory-tracking parameters, and, for $k \in \mathbb{N}$,

$$\beta_k := x(t_k) - \Pi\omega(s_{\theta_{k-1}^*}(t_k)),$$

where x is the state obtained with the path-following policy (20), (21). Let also $V(\beta) := \beta^\intercal P \beta$.

Step 1 (cost of the base option). Consider any $\bar{\theta} \in \Theta_k$ corresponding to constant speed v_c for $t \ge t_k$ and let $\bar{\beta} = x(t_k) - \Pi\omega(s_{\bar{\theta}}(t_k))$. Then $\dot{\beta} = A_K\beta$ between jumps on $[t_k, \infty)$ (see the proof of Proposition 1) and (24) holds with $\hat{t}$ replaced by t_k. Hence

$$\begin{aligned}J_{\mathrm{TT}}^{0,\tau}(x(t_k), t_k, \bar{\theta}) &= \bar{\beta}^\intercal \Big(\int_0^\tau N(r) dr + e^{A_K^\intercal \tau} P e^{A_K \tau} \Big) \bar{\beta} \\ &\quad + \lambda \int_0^\infty \min(r, \tau)\,\mathrm{tr}(N(r)W) dr \\ &\quad + \lambda \int_0^\tau (\tau - r)\,\mathrm{tr}(N(r)W) dr \\ &= V(\bar{\beta}) + \tau\lambda\,\mathrm{tr}(PW),\end{aligned}$$

where we used $e^{A_K^\intercal r} P e^{A_K r} = \int_r^\infty N(q) dq$ and $P = \int_0^\infty N(r) dr$. The second and third terms correspond, respectively, to the noise contributions of the terminal cost and of the running cost.

Step 2 (one-step inequality). By Assumption 5, θ^*_{k-1} has speed v_c on $[t_{k-1}+\tau,\infty) = [t_k,\infty)$, and Θ_k contains the parameters $\hat{\theta}_k$ of the constant-speed trajectory starting at $s_{\theta^*_{k-1}}(t_k)$; thus $r(t,\hat{\theta}_k) = r(t,\theta^*_{k-1})$ for $t \geq t_k$. For $k=0$, let $\hat{\theta}_0 = \theta_0 \in \Theta_0$. In both cases $x(t_k) - \Pi\omega(s_{\hat{\theta}_k}(t_k)) = \beta_k$. On $[t_k, t_{k+1})$, the path-following policy coincides with $\mu_{\mathrm{TT}}(\cdot,\cdot,\theta^*_k)$ and $r(t,\bar{\theta}(t)) = r(t,\theta^*_k)$; moreover, given $\mathcal{F}_k$, the jump times in (t_k,∞) form a Poisson process with rate λ, and $x(t_{k+1}) - \Pi\omega(s_{\theta^*_k}(t_{k+1})) = \beta_{k+1}$. Therefore, using (20) and Step 1,

$$\begin{aligned}
&\mathbb{E}\Big[\int_{t_k}^{t_{k+1}} \|z(t) - r(t,\bar{\theta}(t))\|^2 dt + V(\beta_{k+1})\Big|\mathcal{F}_k\Big] \\
&= J^{0,\tau}_{\mathrm{TT}}(x(t_k),t_k,\theta^*_k) \leq J^{0,\tau}_{\mathrm{TT}}(x(t_k),t_k,\hat{\theta}_k) \\
&= V(\beta_k) + \tau\lambda\,\mathrm{tr}(PW).
\end{aligned}$$

Step 3 (averaging). Taking expectations (which, by induction, are finite), summing for $k \in \{0,\dots,N-1\}$, and using $V(\beta_N) \geq 0$ and Proposition 1,

$$\mathbb{E}\Big[\int_{t_0}^{t_N} \|z(t) - r(t,\bar{\theta}(t))\|^2 dt\Big] \leq V(\beta_0) + N\tau J_{\mathrm{TT,av}}.$$

For $T>0$, let $N = \lceil T/\tau \rceil$. Since the integrand is nonnegative,

$$\frac{1}{T}\int_{\underline{t}}^{\underline{t}+T} \mathbb{E}[\|z(t) - r(t,\bar{\theta}(t))\|^2] dt \leq \frac{V(\beta_0)}{T} + \frac{\lceil T/\tau\rceil\tau}{T} J_{\mathrm{TT,av}},$$

and taking the $\limsup$ as $T\to\infty$ yields $J_{\mathrm{PF,av}} \leq J_{\mathrm{TT,av}}$. □

Trajectory-tracking controller unicycle robot

Let the error coordinates rotated to the robot's body axes be given by

$$\begin{bmatrix}\bar{e}_x(t)\\ \bar{e}_y(t)\end{bmatrix} = \begin{bmatrix}\cos(\varphi(t)) & \sin(\varphi(t))\\ -\sin(\varphi(t)) & \cos(\varphi(t))\end{bmatrix}\begin{bmatrix}e_x(t)\\ e_y(t)\end{bmatrix}.$$

The trajectory-tracking control law is then given as

$$\begin{aligned}
V(t) &= \bar{V}(1-e_\rho(t)) - \zeta_1\bar{e}_x(t)\\
\omega(t) &= \bar{\omega}(t) - \zeta_2\bar{V}\bar{e}_y(t) - \zeta_3 h(\rho(t),\bar{\rho}(t))
\end{aligned}$$

where

$$h(\rho(t),\bar{\rho}(t)) = \begin{cases} -\rho(t)^\top\Gamma^\top\bar{\rho}(t) \text{ if } \rho(t)^\top\bar{\rho}(t)\geq 0;\\ 1 \text{ if } \rho(t)^\top\bar{\rho}(t)<0 \text{ and } \rho(t)^\top\Gamma^\top\bar{\rho}(t)\leq 0;\\ -1 \text{ if } \rho(t)^\top\bar{\rho}(t)<0 \text{ and } \rho(t)^\top\Gamma^\top\bar{\rho}(t)>0.\end{cases}$$

and $\Gamma = \begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}$. ζ_1, ζ_2 and ζ_3 are positive but otherwise arbitrary control gains. By using $\bar{e}(t) = \begin{bmatrix}\bar{e}_x(t) & \bar{e}_y(t)\end{bmatrix}^\top$, and the fact that $\Gamma^\top = -\Gamma$, it can be established that

$$\dot{\bar{e}}(t) = \begin{bmatrix}0 & \omega(t)\\ -\omega(t) & 0\end{bmatrix}\bar{e}(t) + \begin{bmatrix}V(t) - \bar{V}\rho(t)^\top\bar{\rho}(t)\\ \bar{V}\rho(t)^\top\Gamma\bar{\rho}(t)\end{bmatrix}.$$

Let $q(t) := \rho(t)^\top\Gamma^\top\bar{\rho}(t)$ and consider the Lyapunov candidate function $\mathcal{V}(t) = \bar{e}_x^2(t) + \bar{e}_y^2(t) + \frac{2}{\zeta_2}e_\rho(t)$. Using $\dot{\rho} = \omega\Gamma\rho$, $\dot{\bar{\rho}} = \bar{\omega}\Gamma\bar{\rho}$, we have $\dot{e}_\rho = (\bar{\omega}-\omega)q$ and $\dot{q} = (\bar{\omega}-\omega)\rho^\top\bar{\rho}$. Taking the time derivative of $\mathcal{V}$ and plugging in the control law, we obtain

$$\dot{\mathcal{V}}(t) = -2\zeta_1\bar{e}_x^2(t) + 2\frac{\zeta_3}{\zeta_2}h(\rho(t),\bar{\rho}(t))q(t) \leq 0,$$

since $hq = -q^2$ if $\rho^\top\bar{\rho}\geq 0$ and $hq = -|q|$ otherwise. Thus $\mathcal{V}$ is nonincreasing, $\bar{e}$ is bounded, and $\int_{t_0}^{\infty}\bar{e}_x^2(t) + |h(t)q(t)|\,dt \leq \frac{\zeta_2}{2\min(\zeta_1\zeta_2,\zeta_3)}\mathcal{V}(t_0)$. Assume that $\bar{V}>0$ is constant and that $\bar{\omega}$ and $\dot{\bar{\omega}}$ are bounded (bounded curvature of the path and of its derivative). Then ω, $\dot{\bar{e}}$ and $\dot{q}$ are bounded, so $\bar{e}_x^2$ and hq (which is Lipschitz in $(\rho,\bar{\rho})$) are uniformly continuous, and Barbalat's lemma yields $\bar{e}_x(t)\to 0$ and $q(t)\to 0$. Suppose now that $\mathcal{V}(t_0) < 4/\zeta_2$. Then $e_\rho(t) \leq \frac{\zeta_2}{2}\mathcal{V}(t_0) < 2$ for every t, i.e., $\rho^\top\bar{\rho}$ is bounded away from -1, and $q\to 0$ implies $\rho^\top\bar{\rho}\to 1$, i.e., $e_\rho\to 0$. Hence, after some finite time, $h=-q$, and $\ddot{q}$ is bounded. Applying Barbalat's lemma to q gives $\dot{q}\to 0$, i.e., $\omega-\bar{\omega} = -\zeta_2\bar{V}\bar{e}_y + \zeta_3 q \to 0$, which implies $\bar{e}_y\to 0$. Therefore $\bar{e}(t)\to 0$ and $e_\rho(t)\to 0$ as $t\to\infty$, which implies $e(t)\to 0$ as $t\to\infty$.

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